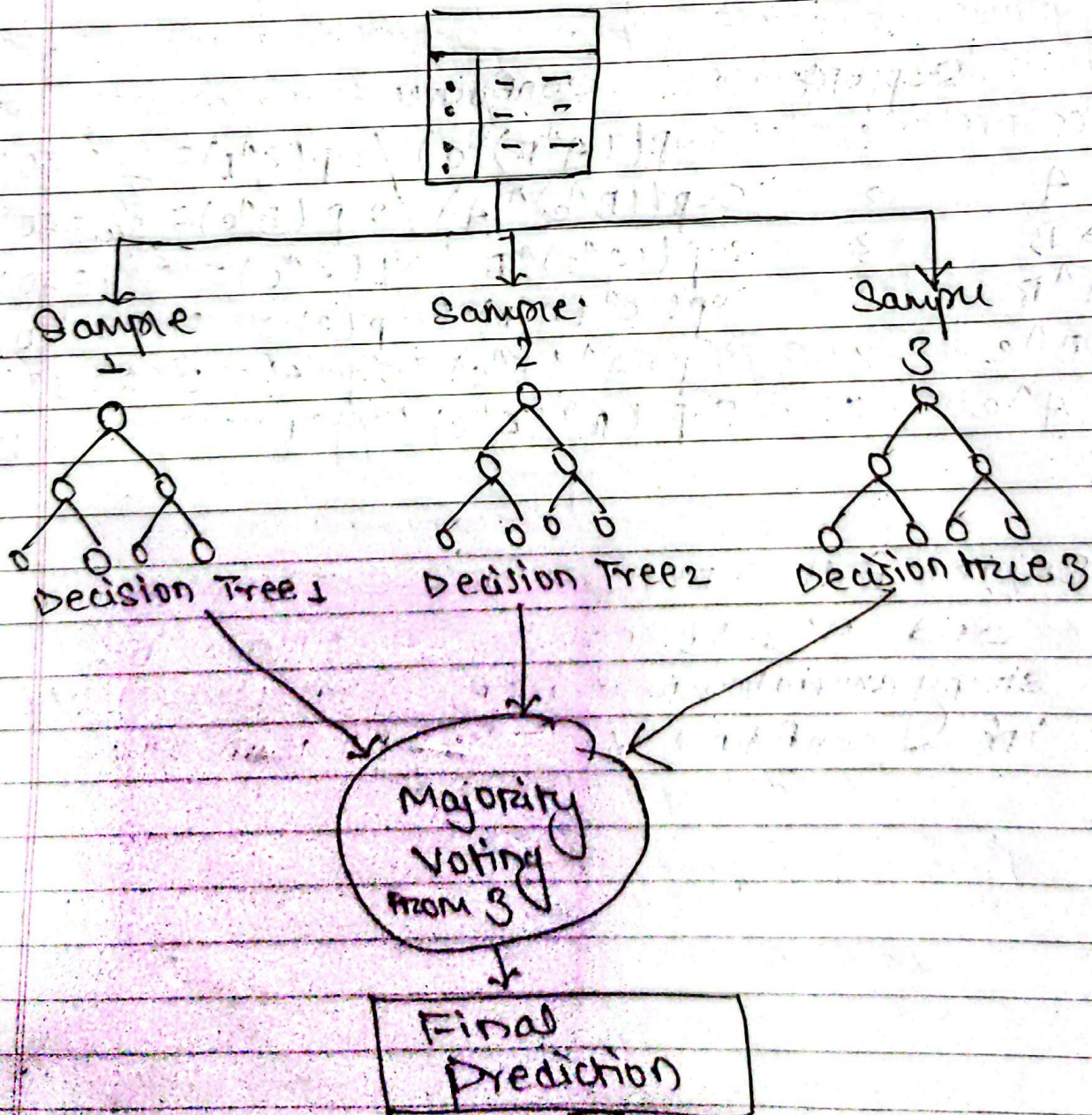


Bagging Vs Boosting

- 1) Bagging : It means training many models independently on different random samples of the data and then combining their result

Bagging Process



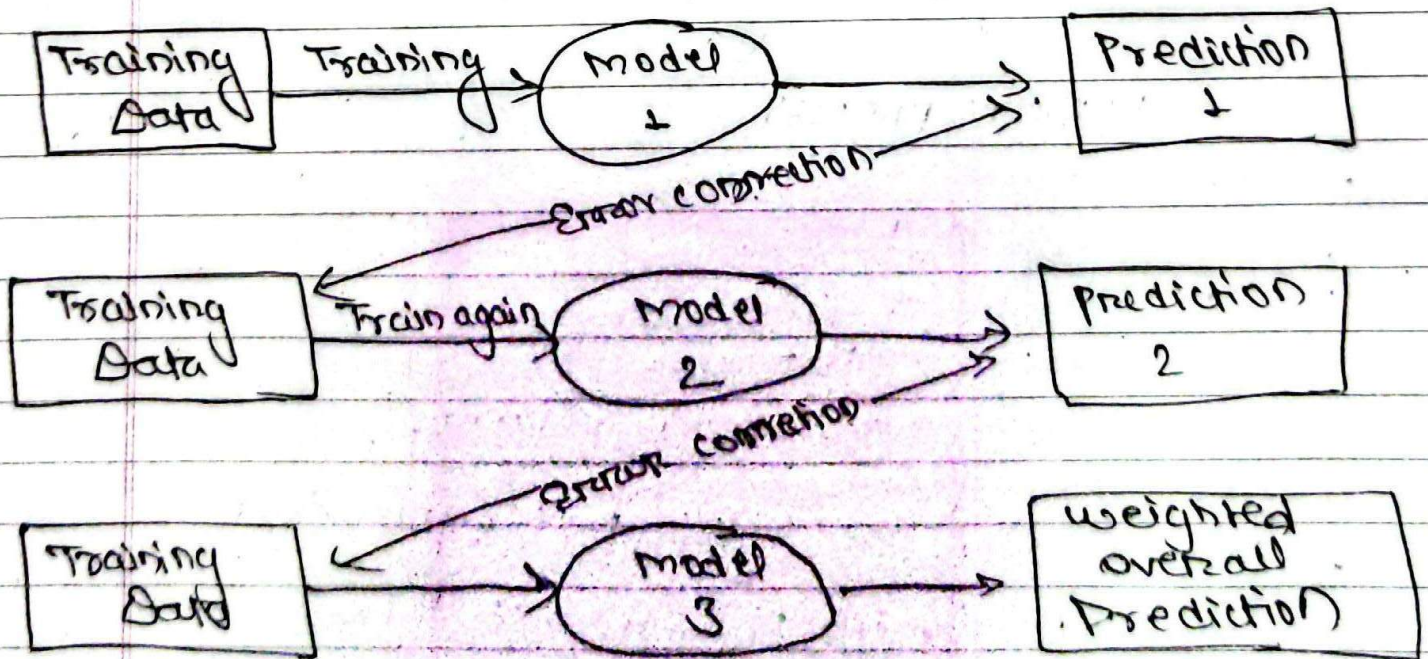
Example:

If 3 trees predict cat, Dog and cat then since the majority of cat is 2 so Final answer will be cat.

→ It's main purpose is to reduce overfitting and variance
eg: Random Forest.

2) Boosting: It means training models one after another where each new model tries to correct the mistake of the previous model.

Boosting Process



→ main purpose is to improve accuracy and reduce bias.
→ eg: Adaboost, Gradient Boosting, XGBoost

Comparing Bagging Vs Boosting

Bagging

- Model work Independently.
- Model is trained parallelly.
- Focus on reducing variance.
- Use different sample.

Boosting

- Model work sequentially.
- Model depend on previous model.
- Focus on reducing ~~error~~ bias.
- New model focus on previous errors.

- 1) AdaBoost & it combine multiple weak models into single strong classifier.

working process

Data



Model 1



Find wrong predictions



Give wrong point more weight



Model 2 focus on them



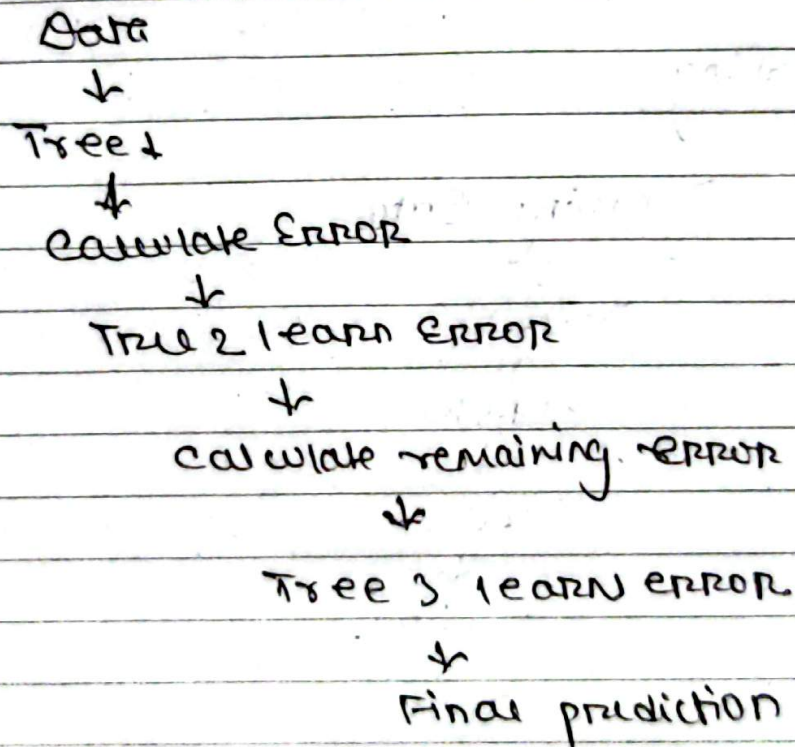
repeat



Final prediction

2) Gradient Boosting & it builds model sequentially to correct previous errors.

working process



eg:

Actual = 100

First tree predict : 80

error = 20

The Next tree tries to learn this 20 error.

3) XG Boosting is an optimised version of Gradient ~~and~~ Boosting that provide faster and more accurate performance.

Working

Training Data



Tree 1



Errors



Tree 2



Errors



Tree 3



regularization + optimization



Final prediction